

Skills Summary

Input, Output, and Function Notation

Use this reference while completing your worksheet or when studying.

Skill 1 — Telling the input from the output

Rule: in $f(a) = b$, the number in the brackets is the **input** and the number after the equals sign is the **output**.

“Find $f(4)$ ” hands you the input and asks for the output. “Find x when $f(x) = 15$ ” hands you the output and asks for the input. Read the question for which one is missing.

Example: $f(x) = 5x + 1$. Find $f(4)$.

Step 1. The 4 sits in the brackets, so it is the input — substitute it and simplify.

Step 2. $f(4) = 5(4) + 1 = 20 + 1 \Rightarrow f(4) = 21$

Try it: $f(x) = 5x + 1$. Find $f(-1)$.

check: $\rightarrow$

Skill 2 — Working backwards from an output

Rule: when the output is given, set the rule *equal* to it and solve for the input. Do not substitute the output for x .

$f(x) = \text{given output} \Rightarrow$ solve the equation. Check your answer by substituting it back into the rule.

Example: $f(x) = 2x + 7$. Find x when $f(x) = 15$.

Step 1. 15 is an output, so it belongs on the right of an equation — solve, do not substitute.

Step 2. $2x + 7 = 15$, so $2x = 8$ and $x = 8 \div 2$. Check: $2(4) + 7 = 15 \checkmark \Rightarrow x = 4$

Try it: $f(x) = 2x + 7$. Find x when $f(x) = 21$.

check: $x =$

Skill 3 — Spotting a swapped substitution

Rule: the input replaces x and *only* x . Every other number in the rule stays exactly where it was written.

To check someone’s work, rewrite the rule with brackets where x was — $4() - 9$ — and drop the input into the gap. If their line does not match that shape, the numbers were swapped.

Example: $h(x) = 4x - 9$. A student wrote $h(-2) = 4(-9) - 2 = -38$. Correct it.

Step 1. Compare the student’s line with the shape $4() - 9$: the input and the constant have changed places.

Step 2. Put the input in the gap: $h(-2) = 4(-2) - 9 = -8 - 9 \Rightarrow h(-2) = -17$

Try it: $h(x) = 4x - 9$. A student wrote $h(3) = 4(-9) - 3 = -39$. Correct it.

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(!) Watch out: an output can look like an input. In $p(5)$ the 5 is an input; in $p(x) = 5$ the same 5 is an output. Check which side of the equals sign it is on.